

# Pratyush Singh

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## SUMMARY

Computer Science Engineering student (B.Tech, 2027) specializing in Cloud Computing and Automation, with a focus on backend and cloud engineering. Works across the stack with TypeScript, Node.js, React, and PostgreSQL, and deploys applications on AWS, Vercel, and Neon. Skilled in REST API design, JWT authentication, role-based access control, real-time WebSocket services, and LLM integration. Microsoft Azure Data Fundamentals certified, with 300+ DSA problems completed on LeetCode.

## EDUCATION

### VIT Bhopal University

B.Tech in Computer Science and Engineering (Cloud Computing and Automation)  
CGPA: 8.00/10

Bhopal, Madhya Pradesh

May 2027

### Bhavan's Kesari Devi Kanoria Vidya Mandir

Class-XII: 73.5%  
Class-X: 92.5%

Sonebhadra, Uttar Pradesh

2023

2021

## TECHNICAL SKILLS

**Languages** : Python, C++, JavaScript, TypeScript, SQL, HTML, CSS

**Frameworks** : React.js, Next.js, Node.js, Express.js, Flask, FastAPI

**Databases** : PostgreSQL, MongoDB, MySQL, SQLite

**Cloud & DevOps** : AWS (EC2), Docker, Git, GitHub, Cloud Deployment (Vercel, Neon)

## PROJECTS

**WorkNest** | Next.js, TypeScript, Node.js, Express, PostgreSQL, Prisma, Socket.IO, JWT

Oct 2025 – Present

- Architected and deployed a multi-tenant cloud SaaS workspace unifying HR, project delivery, and a client portal for service companies, exposing 85+ REST API endpoints across 18 Express modules on a 23-model PostgreSQL schema via Prisma.
- Implemented JWT authentication in httpOnly cookies with bcrypt hashing and role-based access control (RBAC) for 4 roles (Admin, Manager, Employee, Client), isolating tenants by scoping every query to a company ID.
- Secured onboarding with single-use SHA-256-hashed invite and password-reset tokens, brute-force rate limiting (15-minute lockout after 8 failed sign-ins), Zod schema validation, and searchable audit logs.
- Orchestrated a real-time layer with Socket.IO over cookie-authenticated WebSockets and 4 isolated room scopes (company, project, client, user), live-syncing a Kanban board, activity feeds, attendance, and notifications.
- Integrated geofenced attendance (Haversine distance), approval workflows for 4 leave types, and an AI assistant using LLM tool calling (Gemini/Groq); Node.js/Express backend deployed on AWS EC2, with Vercel and Neon PostgreSQL.
- Documented architecture, setup, and deployment in a detailed README, enabling one-command local setup with an embedded PostgreSQL database and no Docker dependency.

**Wardrobe AI** | Next.js, React, TypeScript, Gemini API, Claude API, SQLite, Zod, Tailwind CSS

Oct 2026

- Developed an AI-powered outfit assistant that analyzes an uploaded outfit photo with a multimodal vision LLM and returns a 0–100 suitability score across 4 dimensions: formality, coordination, weather, and event context.
- Engineered a provider-agnostic cloud AI layer integrating 2 LLM providers (Google Gemini and Anthropic Claude) with Zod-validated structured JSON output, ensuring type-safe responses and consistent error handling.
- Created a digital closet with automatic AI tagging across 8 clothing categories, 4 seasons, and 7 occasions, plus outfit recommendations with a rule-based fallback that keeps suggestions working when the AI is unavailable.
- Designed 5 REST API routes with the Next.js App Router and SQLite, integrated live weather from the cloud-based Open-Meteo API, and added client-side image compression to keep photo uploads under the 5 MB limit.

## CERTIFICATIONS

NPTEL: Introduction to Internet of Things Certificate

Microsoft: Azure Data Fundamentals Certificate

## ACHIEVEMENTS

- Earned a 5-Star rating in C++ on HackerRank by solving challenges across Data Structures and Algorithms.
- Solved 300+ coding problems on LeetCode and other platforms, reaching a global LeetCode ranking under 500,000.